3.1.2 Performance characteristics of materials

Performance characteristics of papers and boards

<table>
<tr><th>Content</th><th>Potential links to maths and science</th></tr>
<tr><td>
Students should be able to name different types of papers and boards.
Students should be able to describe the performance characteristics of papers and boards, including:

the ability to be scored
cutting
folding
surface qualities for printing
impact resistance
recyclability and/or biodegradability.

Students should be able to explain why different papers and boards are suitable for different applications, including:

layout paper: sketch pads
cartridge paper: printing
tracing paper: copying images
bleed proof paper: marker rendering
treated paper: photographic printing
watercolour paper: painting
corrugated card: packaging
bleached card: greeting cards and high quality packaging
mount board: modelling
duplex card: food packaging
foil backed and laminated card: drinks packaging
metal effect card: gift packaging
moulded paper pulp: eco-friendly packaging.

</td><td>
Efficient use of materials in the construction of containers through 2D net design.
Effective selection of materials to allow for recyclability, biodegradability and stability.
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</table>

Performance characteristics of polymer based sheet and film

Content	Potential links to maths and science
Students should be able to name different types of polymer based sheet and film. Students should be able to describe the performance characteristics of polymer based sheet and film, including: • the ability to be scored • cutting • folding • moulding • transparency • translucency • flexibility • recyclability and/or biodegradability.	
Students should be able to explain why different polymer based sheet and film are suitable for different applications, including: • foam board: model making • fluted polypropylene: signs and box construction • translucent polypropylene sheets: packaging • styrofoam: modelling and formers • low density polyethylene sheet: wrapping, packaging and bags • plastazote foam: protective packaging • cellulose acetate: packaging • polyactide sheet and film: biodegradable packaging.	

Performance characteristics of woods

Content	Potential links to maths and science
Students should be aware of the different stock forms of timber, including: • rough sawn • planed square edge (PSE) • planed all round (PAR) • natural timber • manufactured boards • mouldings.	

Content	Potential links to maths and science
Students should be able to describe the performance characteristics of woods, including: • grain pattern • grain direction • surface defects • warpage • shrinkage • splitting • joining • forming • steam bending • laminating • machining qualities • resistance to decay • moisture resistance • toxicity.	

Students should be familiar with the following woods and wood products:
• softwoods:
  • pine
  • spruce
  • Douglas fir
  • redwood
  • cedar
  • larch
• hardwoods:
  • oak
  • ash
  • mahogany
  • teak
  • birch
  • beech
• manufactured boards:
  • plywood
  • marine plywood
  • aeroply
  • flexible plywood
  • chipboard
  • medium density fibreboard (MDF)
• veneers and melamine formaldehyde laminates.	

Performance characteristics of metals

<table>
<tr><th>Content</th><th>Potential links to maths and science</th></tr>
<tr><td>Students should be aware of the different stock forms of metals, including:

sheet
plate
bar:
flat
round
square
hexagonal
tube:
round
square
rectangular
hexagonal
structural:
H beam
I beam
tee
channel
angle.
</td><td></td></tr>
<tr><td>Students should be able to describe the performance characteristics of metals, including:

hardness
toughness
malleability
elasticity
tensile strength
density
resistance to corrosion
thermal conductivity
electrical conductivity
melting points
ability to be alloyed
ability to be joined with heat processes
ability to take applied coatings and finishes.
</td><td></td></tr>
</table>

Content	Potential links to maths and science

Students should be familiar with the following metals:
• ferrous:
  • low carbon steel
  • stainless steel
  • high speed steel (HSS)
  • medium carbon steel
  • cast iron
• non-ferrous:
  • aluminium
  • copper
  • zinc
  • silver
  • gold
  • titanium
  • tin
• ferrous alloys:
  • stainless steel
  • die steel (tool steel)
• non-ferrous alloys:
  • bronze
  • brass
  • duralumin
  • pewter.	

Performance characteristics of polymers

Content	Potential links to maths and science
Students should be aware of the different stock forms of polymers, including: • sheet • film • granules • rod and other extruded forms • foam • powder.	

Content	Potential links to maths and science
Students should be able to describe the performance characteristics of polymers, including: • toughness • elasticity • insulation (thermal and electrical) • UV resistance • ability to be moulded • resistance to chemicals and liquids • melting points • suitability for food packaging applications • biodegradability • recyclability • self finishing • ability to be combined with other polymers and/or additives.	

Students should be familiar with the following polymers:
• thermoplastic:
    • low density polyethylene (LDPE)
    • high density polyethylene (HDPE)
    • polypropylene (PP)
    • high impact polystyrene (HIPS)
    • acrylonitrile butadiene styrene (ABS)
    • polymethylmethacrylate (PMMA)
    • nylon
    • rigid and flexible polyvinyl chloride (PVC)
    • Polyethylene terephthalate (PET)
• thermosets, with specific reference to their:
    • urea formaldehyde (UF)
    • melamine formaldehyde (MF)
    • polyester resin
    • epoxy resin.	

Elastomers

Content	Potential links to maths and science
Students should be able to explain the suitability of elastomers for given applications making reference to relevant physical and/or mechanical properties, including: • ability to be stretched and then return to original shape • texture • self finishing • non-toxic. Students should understand how elastomers are used to enhance products, for example in producing grips for improved ergonomics.	
Students should be familiar with the following elastomers: • natural rubber • polybutadiene • neoprene • silicone • Thermoplastic Elastomer (TPE).	

Biodegradable polymers

Content	Potential links to maths and science
Students should be able to explain the suitability of biodegradable polymers for given application making reference to relevant physical and/or mechanical properties, including: • ability to be moulded into 3D products or film • ability to degrade with the action of UV rays (sunlight), water or enzymes present in soil. Students should understand how biodegradable polymers degrade.	

Content	Potential links to maths and science
Students should be familiar with the following biodegradable polymers: • corn starch polymers • potatopak • biopol (bio-batch additive) • polyactide (PLA) • polyhdroxyalkanoate (PHA) • water soluble: lactide, glycolide (Lactel and ecofilm).	

Composites

Content	Potential links to maths and science
Students need to know and understand how materials are combined to make composites with enhanced properties. Students should be able to explain the suitability of composites for given application making reference to relevant physical and/or mechanical properties, including: • ability to be moulded into a variety of 3D forms • enhancement of physical and/or mechanical properties • ease of manufacture for some uses against traditional materials • improved product performance. Students should be familiar with the following composites: • carbon fibre reinforced plastic (CFRP) • glass reinforced plastic (GRP) • tungsten carbide • aluminium composite board • concrete, including reinforced concrete • fibre cement • engineered wood, eg glulam (glued laminated timber).	

Smart materials

Content	Potential links to maths and science
Students should know and understand the term smart material. Students should be able to explain the suitability of smart materials for given applications making reference to how the material responds to external stimuli, including: • changes in temperature • changes in light levels • changes in pressure (force).	
Students should be familiar with the following smart materials: • shape memory alloys (SMA), eg Nitinol • thermochromatic pigment • phosphorescent pigment • photochromic pigment • electroluminescent wire • piezo electric material.	

Modern materials

Content	Potential links to maths and science
Students should know and understand the term modern material.	
Students should be able to explain the suitability of modern materials for given applications.	
Students should be familiar with the following modern materials: • kevlar • precious metal clay (PMC) • high density modelling foam • polymorph.	